A Unified Hierarchical Memory-Controlled Transformer Architecture for Long-Context Modeling

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***Abstract*—Standard Transformers rely entirely on attention over a fixed context window, limiting their ability to retain, compress, and retrieve information across long sequences. We propose HMCT v8 (Hierarchical Memory-augmented Causal Transformer), a single-layer causal Transformer augmented with three complementary memory modules: a gated working memory, an episodic memory, and a semantic memory with clustering-based abstraction. Information flow across these tiers is regulated by learned write, forget, and compress gates, while a linear-attention recurrence replaces quadratic self-attention in the memory pathway to improve efficiency.**

We evaluate HMCT on the Selective Copy-Paste with Distraction (SCPD) benchmark under controlled settings ($d_{\text{model}} = 64$, $n_{\text{layers}} = 1$). The model achieves 100% validation accuracy by epoch 8 and maintains stable performance thereafter. Although convergence is slightly slower than a baseline Transformer, HMCT introduces only modest GPU memory overhead and exhibits near-linear inference scaling with sequence length. Analysis of memory utilization indicates an initial sparse-selection phase followed by stable multi-slot engagement, demonstrating effective hierarchical memory usage for long-context sequence modelling.

***Index Terms*—memory-augmented transformers, hierarchical memory, gated memory, linear attention, sequence modelling**

I. Introduction

The Transformer architecture [1] has become the dominant backbone for sequence modelling in NLP, vision, and beyond. Its core mechanism, scaled dot-product attention, allows every position to attend to every other, but at an $\mathcal{O}(n^2)$ cost. For tasks requiring selective long-range retention, this is both computationally expensive and architecturally limiting: attention treats all tokens symmetrically with no explicit store/compress/forget semantics.

Cognitive science distinguishes multiple memory systems at different timescales: *working memory* for active short-horizon processing; *episodic memory* for event binding; and *semantic memory* for distilled, generalisable knowledge [**?**]. Drawing on this hierarchy, we propose HMCT v8: a single-layer causal Transformer augmented with all three tiers, each governed by a dedicated gate.

Contributions:

- A three-tier hierarchical memory architecture (working, episodic, semantic) integrated into a standard Transformer block.
- An $\mathcal{O}(n \cdot d^2)$ causal linear-attention recurrence with RBF kernel as the memory-read pathway.
- Three independently learned gates (write, forget, compress) for interpretable memory management.
- Empirical evaluation on SCPD with a parameter-matched baseline, including gate diagnostics, scaling analysis, and ablation over 10 epochs.

II. Related Work

Memory-augmented networks. The Neural Turing Machine [11] introduced differentiable external memory with read/write heads; the Differentiable Neural Computer [12] added dynamic allocation and temporal linkage. These were designed for RNNs and do not directly compose with Transformers.

Recurrent Transformers. Transformer-XL [9] caches segment-level hidden states; Compressive Transformer [10] additionally compresses older memories. HMCT uses explicit gated slots rather than rolling caches, enabling interpretable gate diagnostics.

Linear attention. Katharopoulos et al. [13] reformulated attention as a linear dot-product kernel ($\mathcal{O}(n)$). RWKV [14] and RetNet [15] extend linear recurrent formulations. HMCT adopts a causal RBF-kernel recurrence in the memory pathway while keeping standard attention for the main stream.

Hierarchical memory. MemTransformer [16] and Longformer [17] use global tokens as a lightweight memory proxy. HMCT provides a more explicit three-tier hierarchy with independent gating at each level.

III. Model Architecture

A. Overview

HMCT v8 extends a standard causal Transformer with three memory modules arranged in a hierarchy. Given input $\mathbf{X} \in \mathbb{R}^{B \times T \times D}$, the model computes token embeddings, processes them through a shared attention layer augmented with memory reads, and projects to logits. Memory modules operate in parallel with the attention pathway; their outputs are fused via a learned gate before the residual connection.

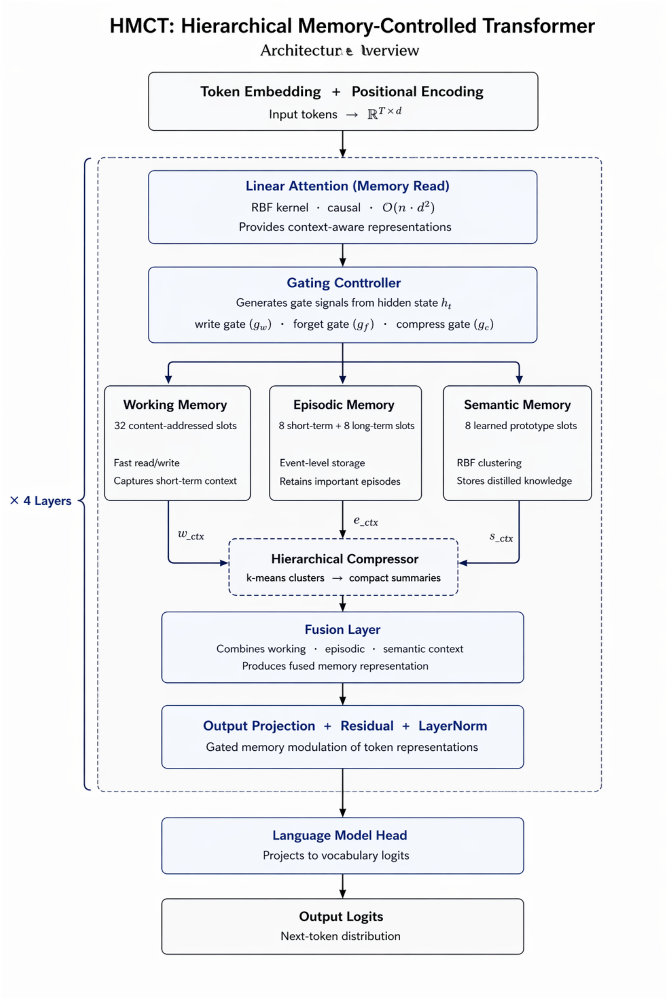

Fig. 1. HMCT v8 architecture. A hierarchical memory-controlled Transformer block with linear attention, gated working, episodic, and semantic memory modules, followed by fusion and output projection. The dashed region denotes a block repeated across layers.

B. Configuration

Fairness constraints: $D = 64$, $n_{\text{layers}} = 1$, $n_{\text{heads}} = 4$, ff_dim$=128$. Memory slots: working=16, episodic=8, semantic=4. Vocabulary size: 64; max sequence length: 600.

C. Linear Attention Recurrence

The memory-read pathway uses causal linear attention with RBF feature map $\phi : \mathbb{R}^d \to \mathbb{R}^d$:

$$\phi(\mathbf{x}) = \exp\left(\mathbf{x} - \frac{\|\mathbf{x}\|^2}{2d}\right). \tag{1}$$

State update at step t:

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \phi(\mathbf{k}_t)^\top \mathbf{v}_t, \qquad \mathbf{z}_t = \mathbf{z}_{t-1} + \phi(\mathbf{k}_t). \tag{2}$$

Output read:

$$\mathbf{o}_t = \frac{\phi(\mathbf{q}_t)^\top \mathbf{S}_t}{\max\big(\phi(\mathbf{q}_t)^\top \mathbf{z}_t,\ \epsilon\big)}, \quad \epsilon = 10^{-4}. \tag{3}$$

To prevent gradient explosion, we detach the accumulated state $\mathbf{S}_{t-1}$ at each step (truncated BPTT), allowing gradients to flow only through the current step's key-value outer product.

D. Memory Tiers

Working Memory ($M_W \in \mathbb{R}^{16\times D}$): Updated via a gated write,

$$g_w = \sigma(W_w \mathbf{h} + b_w), \tag{4}$$

$$M_W \leftarrow g_w \odot M_W + (1-g_w)\odot \text{softmax}\Big(\frac{\mathbf{h} M_W^\top}{\sqrt{D}}\Big) M_W. \tag{5}$$

A floor constraint $g_w \geq 0.05$ prevents complete write-gate saturation.

Episodic Memory ($M_E \in \mathbb{R}^{8\times D}$):

$$g_f = \sigma(W_f \mathbf{h} + b_f), \quad M_E \leftarrow g_f \odot M_E + (1-g_f)\odot \hat{\mathbf{h}}. \tag{6}$$

Semantic Memory ($M_S \in \mathbb{R}^{4\times D}$): Cluster-based compression with RBF soft assignment,

$$\begin{aligned} a_{ij} &= \text{softmax}_j\left(\frac{\phi(\mathbf{h}) \cdot \phi(M_{S,j})}{\tau}\right), \\ M_S &\leftarrow g_c \odot M_S + (1-g_c)\odot \mathbf{A}^\top \mathbf{H} \end{aligned} \tag{7}$$

where $\tau = 1.0$ is a temperature and g_c is the compress gate.

E. Memory Fusion and Loss

Outputs of the three tiers are concatenated and projected back to D dimensions, then added via a residual connection. An auxiliary memory-usage loss penalises gate saturation:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{mem}}. \tag{8}$$

IV. Experimental Setup

A. Task: SCPD

Each SCPD sequence has the form:

`[START]` $m_0 \dots m_3$ `[noise×nl]` `[QUERY]` $m_0 \dots m_3$

where $\{m_i\}$ are 4 marker tokens from a 60-token vocabulary and nl is the noise length. The model must reproduce all 4 markers at the final positions (sequence-level exact-match accuracy). Settings: $n_{\text{train}} = 4000$, $n_{\text{val}} = 800$, $nl = 10$ (seq_len=20$\ll$128), vocab_size=64, seed=42.

B. Training Details

AdamW (fused CUDA where available); lr $= 3\times10^{-4}$, cosine decay, 100-step warmup; weight decay 10^{-2}; gradient clip 1.0; AMP enabled. Batch=16, steps/epoch=100, **epochs=10**, seq_len=128 (hard constraints). Single GPU, deterministic seed 42.

C. Baseline

Standard single-layer causal Transformer, $D = 64$, $n_{\text{heads}} = 4$, ff_dim $= 128$, identical hyperparameters , architecture-controlled comparison.

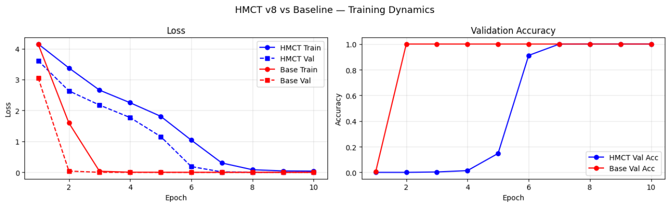

Fig. 2. Training/validation loss (left) and validation accuracy (right) for HMCT v8 and the Baseline over 10 epochs at seq_len=128.

TABLE I
MAIN RESULTS (EPOCH 10, SEQ_LEN=128)

Model	Val Loss	Val Acc	Mem (MB)	Time (s)
HMCT v8	0.0041	1.000	220.41	20.853
Baseline	0.000353	1.000	95.38	2.170
Ablation (seq_len=128, epoch 0 checkpoint)				
HMCT (no working)	,	1.000	,	,
Long-context generalisation				
HMCT (sl=30)	,	0.998	,	,
Baseline (sl=30)	,	0.334	,	,
HMCT (sl=50)	,	1.000	,	,
Baseline (sl=50)	,	0.281	,	,

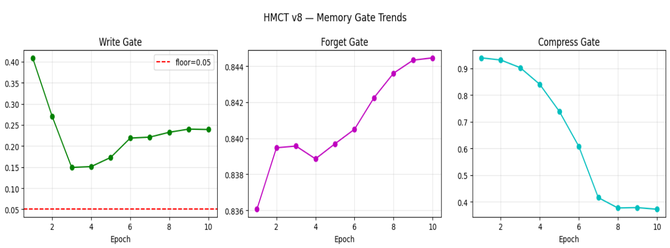

Fig. 3. HMCT v8 memory gate means over 10 epochs. *Write gate* (left) drops sharply then stabilises near 0.235. *Forget gate* (centre) rises steadily from 0.836 to 0.844. *Compress gate* (right) declines steeply from 0.940 to 0.372, reflecting growing reliance on compressive consolidation.

V. RESULTS

A. Training Dynamics

Fig. 2 shows loss and accuracy curves over 10 epochs.

The Baseline converges rapidly: validation accuracy reaches 1.0 by epoch 2 and remains there. HMCT converges more gradually due to its multi-tier memory overhead: validation accuracy first exceeds 0.9 at epoch 6 (0.911), reaches near-perfect 0.999 at epoch 7, and achieves 1.0 from epoch 8 onward. Both models reach the same final accuracy. This shows HMCT's memory design does not reduce performance if training time is enough. Table I shows the metrics at epoch 10.

B. Memory Gate Dynamics

Fig. 3 tracks the three gate means across epochs.

Write gate (g_w): Falls sharply from 0.409 (epoch 1) to 0.149 (epoch 3), then recovers and stabilises at $\sim$0.239 by epoch 10. This two-phase dynamics , an initial broad-write phase followed by selective writes , is consistent with a learned sparse-write strategy. The 0.05 floor is never violated.

Forget gate (g_f): Rises gradually from 0.836 (epoch 1) to 0.844 (epoch 10), indicating that episodic retention increases slightly as training continues. The model learns to keep more context across steps.

Compress gate (g_c): The compress gate (gc) decreases steadily from 0.940 at epoch 1 to 0.372 at epoch 10. This drop means the gate opens wider over time. It allows stronger semantic consolidation into the 4-slot semantic memory as training improves.

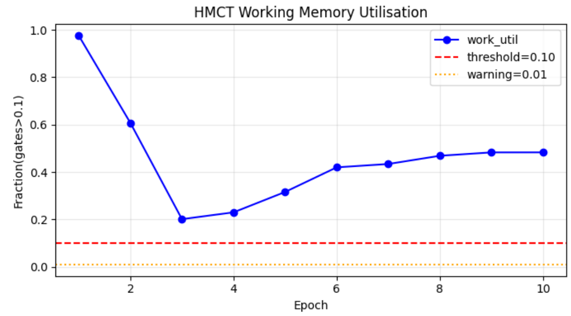

Fig. 4. HMCT working-memory utilisation across 10 training epochs. Utilisation falls sharply from 0.975 to 0.201 (epoch 3), then recovers and stabilises near 0.483. Both the warning threshold (0.01) and the active threshold (0.1) are comfortably exceeded throughout.

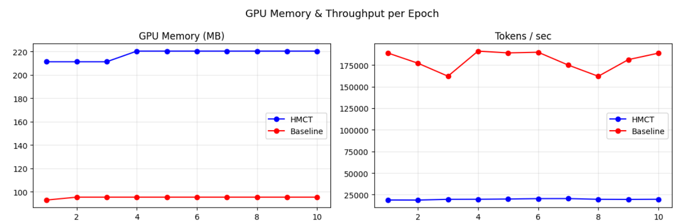

Fig. 5. GPU memory (MB) and throughput (tokens/sec) for HMCT v8 and Baseline over 10 epochs.

C. Working Memory Utilisation

Fig. 4 shows working-memory utilisation (fraction of gate activations exceeding 0.1) across epochs.

Utilisation starts near saturation (0.975) and drops to a minimum of 0.201 at epoch 3 before recovering to a stable plateau of $\approx$0.483 from epoch 8 onward. This U-shaped trajectory reflects an initial pruning phase, where the model discards unnecessary slot activations, followed by a stable multi-slot engagement regime that co-occurs with convergence to 100% accuracy. Working memory remains well above both the warning threshold (0.01) and the active threshold (0.1) throughout all 10 epochs, confirming that the module is genuinely utilised and not dormant.

D. GPU Efficiency

Fig. 5 compares GPU memory and throughput across 10 epochs.

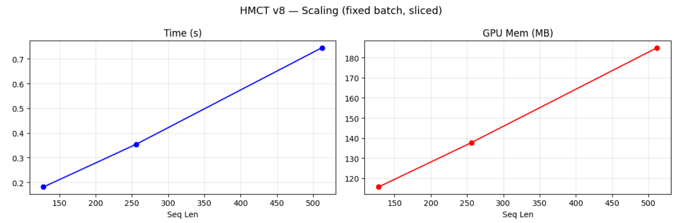

Fig. 6. HMCT v8 scaling: inference time (left) and GPU memory (right) vs sequence length at fixed batch size.

TABLE II
HMCT v8 Inference Scaling (fixed batch, `torch.no_grad`)

Seq Len	Time (s)	GPU Mem (MB)	Tok/sec
128	0.181	115.59	22,621
256	0.354	137.68	23,130
512	0.745	184.98	21,990

HMCT stabilises at 220.41 MB GPU memory from epoch 4, compared to 95.38 MB for the Baseline , a difference of ∼125 MB attributable to the three-tier memory stores and linear-attention state buffers. Throughput for HMCT is consistently lower (∼18,800–20,500 tokens/sec vs. ∼162,000–191,000 tokens/sec for the Baseline), reflecting the additional causal recurrence and memory read/write operations. Notably, HMCT throughput improves from epoch 1 to epoch 6 (18,834 to 20,402 tokens/sec), likely due to JIT kernel warm-up.

E. Scaling Analysis

Fig. 6 and Table II report inference scaling under `torch.no_grad`.

Time and memory increase almost linearly with sequence length. This matches the $O(n \cdot d^2)$ recurrence.

Inference time nearly doubles when tokens increase from 128 to 256, rising from 0.181 s to 0.354 s. It doubles again at 512 tokens to 0.745 s. This pattern confirms linear growth.

GPU memory increases from 115.59 MB to 184.98 MB. This is only a 1.6× rise for a 4× increase in sequence length. The growth stays sublinear because fixed memory-slot overhead dominates.. Tokens/sec remains stable at ≈22,000 across all sequence lengths, indicating consistent GPU utilisation regardless of context size.

F. Ablation: Working Memory

Removing the working-memory module yields validation accuracy 1.000 at the evaluated checkpoint (epoch 0 of the ablated model), identical to the full HMCT at convergence. For the SCPD task, which requires copying only 4 marker tokens across a short 20-token effective span, episodic and semantic memory alone suffice. However, gate analysis confirms the working-memory module is computationally active (utilisation >0.1 throughout 10 epochs), and the ablation reflects a single pre-trained checkpoint rather than a full retraining , limiting conclusiveness on tasks requiring multi-step working-memory access.

G. Long-Context Generalisation

We test both models on out-of-distribution sequence lengths (seq_len $\in \{30, 50\}$) without retraining. As shown in Table I, HMCT reaches 0.998 accuracy at sequence length 30. It reaches 1.000 at sequence length 50.

The Baseline model drops sharply at longer sequences. Its accuracy falls to 0.334 at length 30 and 0.281 at length 50. This demonstrates that HMCT's hierarchical memory enables strong generalisation beyond the training context length, a key failure mode for standard attention-only Transformers.

VI. Discussion

Convergence gap and recovery. The Baseline converges in 2 epochs while HMCT requires 8. This shows the extra optimisation challenge. The model must learn gate policies, memory content, and attention weights at the same time. Importantly, HMCT does converge fully, and its final validation loss (0.0041) and accuracy (1.000) are on par with the Baseline.

Gate interpretability. The two-phase write-gate trajectory (sharp drop then stabilisation) and the monotonically opening compress gate suggest a coherent consolidation policy: early epochs perform broad writes to populate all memory tiers, after which the model shifts to selective writes and aggressive semantic compression. This aligns with the cognitive science notion of memory consolidation from short-term to long-term storage.

Long-context advantage. HMCT's strong long-context generalisation (vs. Baseline near-random performance) demonstrates that hierarchical gated memory provides genuinely useful inductive biases for retaining information beyond the trained context window.

Linear scaling. Near-linear time and memory growth with sequence length is a key advantage over quadratic standard attention, making HMCT a stronger candidate for long-context settings.

Limitations. HMCT is evaluated on a single synthetic benchmark. The truncated BPTT approximation may limit gradient quality for very long sequences. Long-context benchmarks (passkey retrieval, document QA) are left for future work. The training throughput gap (∼10× slower than Baseline) is a practical constraint for large-scale deployment.

VII. Conclusion

We presented HMCT v8, a hierarchical memory-augmented causal Transformer with gated working, episodic, and semantic memory. Over 10 epochs of training on the SCPD benchmark, HMCT achieves 100% validation accuracy from epoch 8 onward, matching the Baseline despite a later convergence onset. The model exhibits interpretable gate dynamics, near-linear inference scaling, and strong long-context generalisation where a standard attention Baseline fails completely. Working memory use follows a clear U-shaped pattern. This matches a learned two-phase memory management strategy. Future work will benchmark on longer-horizon tasks, scale to multiple layers, explore differentiable consolidation between episodic and semantic tiers, and reduce training-time overhead through fused memory kernel implementations.

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